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MILITARY SPECIFICATION

HOOK, HELICOPTER EXTERNAL CARGO, 20,000 POUND CAPACITY, TYPE C

This Specification has been approved by the Bureau of Naval Weapons, Department of the Navy, Washington, D.C.

1. SCOPE

1.1 This Specification establishes the requirements for one type of a 20,000 pound capacity helicopter external cargo hook, designated Type C.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids form a part of this specification. In case of conflict with the requirements of any specification, the detail requirements specified herein shall govern.

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Boxes, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-S-4040	Solenoid, Electrical, General Specification for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-W-6858	Welding; Resistance: Aluminum, Steels or Alloys, Magnesium, Non-Hardening Nickel Alloys, Heat-Resisting Alloys and Titanium Alloys, Spot and Seam

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SPECIFICATIONS (cont'd)

Military

MIL-W-6873	Welding; Flash, Carbon and Alloy Steels
MIL-P-6906	Plates, Information and Identification
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-P-007936	Parts and Equipment, Procedures for Packaging of
MIL-H-7982	Handbooks; Operation, Service and Overhaul Instructions with Parts Breakdown
MIL-A-8064	Actuators and Actuating Systems, Aircraft, Electro-Mechanical, General Requirements for
MIL-W-8604	Welding of Aluminum Alloys, Process for
MIL-W-8611	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for
MIL-A-8625	Anodic-Coatings, for Aluminum and Aluminum Alloys
MIL-D-70327	Drawings, Engineering and Associated Lists
MIL-L-7961	Light, Indicator, Press to Test
WR-30	Weapons Requirements, Integrated Maintenance Management for Aeronautical Weapons, Weapons Systems and Related Equipment

STANDARDS

Military

MIL-STD-838	Lubrication of Military Equipment
MIL-STD-704	Electrical Power, Aircraft Characteristics and Utilization of
MIL-STD-831	Test Reports, Preparation of
MS 3102	Connector, Receptacle, Electric, Box Mounting

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SPECIFICATIONS (contd)

STANDARDS

Military

MS3106	Connector, Plug, Electric, Straight
MS 33586	Metals. Definition of Dissimilar

3. REQUIREMENTS

3.1 Qualification.- The helicopter external cargo hook furnished under this Specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials.- Materials used in the manufacture of the cargo hook shall be of high quality, suitable for the purpose intended and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used, provided the specifications are approved by the Government and contain provisions for adequate tests. The use of contractor's specifications will not constitute a waiver of Government inspection.

3.2.1 Metals.- All metals used in the construction of the cargo hook shall be of the corrosion-resistant type or shall be suitably treated to resist corrosion per Specification MIL-A-8625, QQ-C-320 or QQ-P-416. The use of dissimilar metals in contact, as defined in MS 33586 shall be avoided.

3.2.2 Standard Parts.- Standard parts (AN or MS) shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they are fully equivalent to and replaceable by standard parts without alteration and provided that the corresponding standard part numbers are referenced on the drawings and in the parts list. In the event there is no suitable MS or AN standard part in effect on the date of invitation for bids, commercial parts may be used provided they conform to all requirements of this Specification.

3.3 Design and Construction.- The cargo hook shall provide a helicopter with an external capability for pickup, transport and release of suitable cargo. The hook shall lend itself readily to either automatic or semi-automatic type cargo pickup. Automatic cargo pickups are considered to exist wherein the helicopter pilot flies the aircraft so that the hook engages with the load without requiring the aid of ground personnel to physically engage the hook with the load. Semiautomatic pickups require ground personnel to engage the load sling in the hook, but without having contact with the hook.

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3.3.1 The hook shall feature reliable operation with simplicity of design as described herein. The hook shall be so designed that no special tools shall be required for assembly, disassembly, or routine maintenance.

3.4 Screw Threads.- All screw threads shall be in accordance with Specification MIL-S-7742.

3.5 Threaded Parts.- All threaded parts shall use accepted aircraft locking methods. Lock washers shall not be used for locking.

3.6 Welding. Welding, where used, shall be in general accordance with Specification MIL-W-6858, MIL-W-8611, MIL-W-8604 and MIL-W-6873.

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Change in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.8 Operating Condition.- The cargo hook shall operate satisfactorily under the following environmental conditions.

3.8.1 Altitude.- From sea level to 20,000 feet.

3.8.2 Temperature.- From -65°F to +160°F.

3.8.3 Vibration.- As specified in 4.6.2.9

3.8.4 Sand and Dust and Salt Spray.- As specified in 4.6.2.12 and 4.6.2.13.

3.8.5 Shock.-As specified in 4.6.2.17.

3.8.6 Water Immersion.-Sea water immersion of the external hook shall not result in any permanent damage nor shall operation be impaired. A fresh water washdown is permissible after completion of operations.

3.8.7 Fungus.- As specified in 4.6.2.16.

3.9. Detailed Requirements.-

3.9.1 General.- The hook shall be of the open-throat type, i.e., the load sling adapter ring or other pickup device may be introduced into the load bearing portion of the hook automatically or in any event it shall not require personal contact with the hook. There shall be no protuberances or other obstacles to prevent, or hinder, engagement with the load sling adapter ring or other pickup device. The open-throat shall face forward when installed on an aircraft.

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3.9.1.1 Throat.- The throat shall be that area between the hook housing and the load beam, open at the forward end and culminating in the area where the load beam supports the sling adapter ring while transporting a cargo load. The throat configuration shall facilitate engagement with the load sling adapter or other pickup device. The throat shall be of adequate area and of a configuration to accept any of the following sling adapter rings or pickup devices or their equivalents:

(1) Nylon strap ring - 2-3/4 inch wide and 7/8 inch thick.
(2) A steel ring made of 1-1/2-inch diameter stock.
(3) Two (2) - 2-inch diameter ropes.

3.9.1.2 Housing.- The hook housing shall be constructed to support and enclose the required components and to protect them from shocks and environmental conditions as required. Any components contained therein which are subject to water damage shall be enclosed within a completely waterproof housing or other adequate protective device.

3.9.1.3 Load Beam.- The load beam shall be adequate to support all loads set forth herein under all operational conditions. The load beam shall be mounted in the hook housing by a pivot near its aft end such that upon release the front end will swing downward and aft. The configuration of the load beam shall be such as to assist the load sling adapter ring to satisfactorily position itself in the hook while presenting minimum resistance to disengagement on release.

3.9.1.4 Keeper.- A keeper shall be employed in the hook throat to positively retain the load sling adapter ring in the desired position in the hook under all operational flight conditions until release. The keeper shall permit easy engagement of pickup devices consisting of more than one part, or ring, each engaged separately without personal contact with any part of the hook.

3.9.1.5 Suspension.- The hook shall have two attachment holes on its upper portion, one near the forward end and one near the aft end as shown in Figure 1, such that the point at which the cargo load is applied is approximately midway between them. These suspension holes shall be adequate for attachment to a semi-rigid mounting structure, as defined in paragraph 6.3, beneath the aircraft and shall permit easy removal of the hook when desired.

3.9.1.6 Release Mechanism.- The internal release mechanism in the hook shall provide positive retention of the load beam in its load carrying position until the release mechanism is actuated by one of the modes of release defined herein. Upon actuation, the release mechanism shall react so as to remove all restraint from the load beam. The load beam shall then be capable of opening and completely releasing the specified loads. Components of the release mechanism which rely on a wiping or rolling action between members shall be self-cleaning to prevent contamination and fouling.

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3.9.2 Weight.- The maximum weight of the hook shall not exceed 40 pounds.

3.9.3 Envelope.- The overall dimensions of the hook shall not exceed those shown in Figure 1.

3.9.4 Capacity.- The Type C cargo hook shall have a 20,000 pound design load rating.

3.9.4.1 The hook shall operate satisfactorily under loads as specified herein, at attitudes from horizontal to a 30 degree nose pitch-up. Application of a torsional moment to the hook while under rated load, as specified herein, shall cause no damage to the hook.

3.9.5 Design Load Factors.- The hook shall be capable of withstanding a design limit load (static) of 60,000 pounds with no yield or damage to working parts, and a design ultimate load (static) of 90,000 pounds.

3.9.6 Operating Features.-

3.9.6.1 Hook Release.- The cargo hook shall provide the following methods of releasing the load from the hook. Actuation of any of the releases for a continuous period of thirty (30) seconds shall not result in any damage to the hook or its components.

3.9.6.1.1 Normal Release.- The normal release circuit shall be actuated by application of the required voltage to it as initiated by a "Normal Release" switch within the aircraft. The device which actuates the release mechanism for normal release shall be capable of releasing all loads, within a range of at least 7 pounds to 6000 pounds in not more than one-half (1/2) of one second from the moment of initiation to release of the cargo. The hook release device can be either an electrical solenoid, per Specification MIL-S-4040, or an electrical actuator, per Specification MIL-A-8064.

3.9.6.1.2 Automatic Touchdown Release.- Automatic touchdown release shall be provided so that when the load on the hook is reduced to 150 ±25 pounds by ground contact, the hook shall release any load down to at least 7 pounds and remain in the released condition until the circuit is de-activated. The actual physical release mechanism shall utilize the normal release mechanism components insofar as possible, with the actual initiating command being by means of a separate mechanism or device, which does not necessarily have to be electrical in operation. The circuits shall be such that this mode of release can be selected by the pilot by means of an electrical switch. The normal release mode must not be deactivated when the automatic touchdown mode is used.

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3.9.6.1.3 Manual Release.- The cargo hook shall contain a mechanical release which will provide a means for manual release of the hook from a remote position when connected to a suitable linkage, or cable, installed in the helicopter. This release shall be capable of releasing any loads within a range of at least 7 pounds up to the design limit of 60,000 pounds. The hook need not be operable after release of loads between the rated capacity and the design limit load but shall be capable of being returned to full operating condition with only minor repairs or replacement of minor parts. The hook and its components shall incur no damage from releases of loads up to rated capacity in this mode. The stroke through which the actuating linkage must travel to execute a complete release of the hook shall not be less than one-half (1/2) inch nor exceed one and one-quarter (1-1/4) inches including any pretravel. The amount of pretravel shall not exceed the actuating travel. The point at which the actuating force is applied to the mechanism shall be so located that the mechanism shall properly release when a connection is made between it and a vertically arising point located approximately four inches above the mounting holes, two inches to starboard of the fore-and-aft hook centerline, and not more than two inches forward or aft of the mid point between the mounting holes. The configuration of the manual release attachment point shall be such that it will accomodate a standard AN or MS terminal at the end of a one-sixteenth (1/16) inch diameter cable The force required to actuate this release under various load conditions shall not exceed the following:

LOAD	FORCE
None	15 lbs.
20,000 lbs.	50 lbs.
60,000 lbs.	80 lbs.

The release mechanism shall contain a means of providing a force of 9 ± 1 pounds at the point of attachment of the actuating linkage in a direction which will tend to retain the aircraft linkage in its de-actuated position. This force is included in the release forces listed above.

3.9.6.1.4 Ground Release.- There shall be a manual means of release provided on the hook for convenient use of ground personnel. The force required to actuate the release mechanism of the hook by this method shall be a maximum of fifteen (15) pounds under no-load condition. The release mechanism shall be such that falling or swinging objects, such as cables or heavy weights, shall not inadvertently actuate the release mechanism.

3.9.6.2 Relatch.- After release of any load up to rated capacity, the hook load beam shall automatically relatch into its load carrying condition within 5 seconds after the release controls have been returned to the non-release position.

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In the event the load beam does not relatch after release of a load, reactuation of either the normal or manual release shall be sufficient action to cause the load beam to relatch.

3.9.6.3 Indication of Unlatched Load Beam.- There shall be a means provided in the hook to indicate when the load beam is in an unlatched condition. The device and circuitry used shall, when this condition is existent, provide a closed circuit of sufficient integrity to permit an indicator light, per Specification MIL-L-7961, to be illuminated in the aircraft when the operating voltage is impressed on the overall circuit.

3.9.7 Electrical Characteristics.-

3.9.7.1 Electric Power.- The hook shall be designed to provide satisfactory operation when furnished, through appropriate control circuit, with the following electric power as defined in MIL-STD-704: 28 Volt DC, Category C, 15 amperes.

3.9.7.2 Low Input Voltage.- The hook shall be capable of effecting a normal release of all loads within a range of at least 7 pounds to 6000 pounds at an input voltage of 22 VDC.

3.9.8 Electrical Connections.-

3.9.8.1 General.- An electrical receptacle, MS3102E14S-5P, shall be provided on the hook for interconnection of the hook electrical circuits with those in the helicopter. The following pin schedule shall be used:

Pin Designation	Function
A	Automatic Touchdown Arming
B	Normal Release
C	Ground
D	Load Beam Unlatched Indication
E	Spare

The receptacle shall be positioned to permit entry of the mating MS3106E-14-5S plug from one of the following directions; top or forward end, without the plug and related hardware obstructing the mounting provisions or projecting outside the dimensional envelope shown in Figure 1.

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3.10 Lubrication.- The hook and all components shall operate satisfactorily with minimal lubrication as defined in 3.11.3. Excessive lubrication shall not contribute to fouling due to entrapment of foreign matter. Any provisions for lubrication shall be in accordance with MIL-STD-838.

3.11 Maintenance.-

3.11.1 Special Training.- The hook shall be such that maintenance personnel shall not require special training.

3.11.2 Time Required for Maintenance.- The hook shall be designed so as to minimize maintenance time. Time required for disassembly, inspection, cleaning and reassembly of the hook shall not exceed three (3) man-hours.

3.11.3 Period.- The hook shall be designed so as to provide satisfactory operation with the following maintenance inspection periods:-

a. A sight inspection and operational check in the normal, manual, and automatic release conditions every 60 hours of aircraft time, or 60 days, whichever occurs first. Hook loadings need only be sufficient to provide a reliable indication of proper operation.

b. Inspection, lubrication if required, and full-load operational check in all operating modes every 240 hours of aircraft time, or six (6) months, whichever occurs first.

c. Complete disassembly and overhaul every 5,000 cycles of hook operation or two (2) years, whichever occurs first.

3.12 Nameplate.- A nameplate, giving the following information, shall be provided and securely attached to the hook:-

Hook, Helicopter External Cargo, Type C
Rated Capacity: 20,000 pounds
Stock Number:_____
Name of Manufacturer:_____
Manufacturer's Part No.:_____
Manufacturer's Serial No.:_____
Contract No.:_____
Specification No. MIL-H-81014 (WEP) U. S. Property

The nameplate shall be in accordance with Specification MIL-P-6906.

3.13 Instruction Plate.- A suitable instruction plate shall be attached to the hook depicting the schematic wiring diagram of the hook.

3.14 Handbook.- When specified in the contract or purchase order, a Handbook of Operating and Maintenance Instructions, including Parts Breakdown shall be prepared in accordance with Specification MIL-H-7982.

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3.15 Workmanship.- Workmanship and construction shall be to the highest aircraft standards and shall reflect the work of skilled craftsmen.

3.16 Integrated Maintenance Management Information.- If required by the contract, the contractor shall supply to the procuring agency, or to the aircraft prime contractor if so directed in the contract or purchase order, all information required for the preparation of Maintenance Engineering Analysis Requirements (MEAR's) and to meet the other requirements of Specification WR-30 for fulfillment of the Integrated Maintenance Management Program for the hook system when installed in a helicopter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification when such inspections are deemed necessary to assure supplies and services that conform to prescribed requirements. All tests shall be conducted at the contractor's expense unless otherwise specified in the contract or purchase order.

4.1.1 Test Results.- Reports on all tests shall be prepared in accordance with Standard MIL-STD-831 and submitted to the procuring agency. An opportunity shall be given to procuring agency representatives to witness all tests.

4.2 Classification of Tests.- The inspection and testing of the cargo hook shall be classified as follows:-

(1) Qualification Tests (See 4.3)
(2) Acceptance Tests (See 4.4)

4.3 Qualification Tests.-

4.3.1 Qualification Test Sample.- One qualification test sample shall be subjected to the qualification testing. This test sample shall remain the property of the manufacturer unless otherwise noted in the contract or purchase order. If the test sample is to become Government property at the completion of qualification, it shall be returned to as good as new condition prior to delivery. The qualification test sample shall be identical to the production hooks. Samples shall be appropriately identified with the manufacturer's own part number and any additional identification required by the authorizing letter.

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4.3.2 Scope of Qualification Tests.- The qualification test sample shall be subjected to all the tests specified under 4.6 to assure conformance of the hook to the requirements of this Specification.

4.3.3 Qualification by Similarity.- Unless otherwise specified, qualification of cargo hooks may be approved on the basis of their similarity to previously approved units when it can be shown to the complete satisfaction of the procuring agency that such similarity is applicable. The qualification tests under 4.6 may be modified as necessary, upon approval of the procuring agency, when approval by similarity is sought.

4.4 Acceptance Tests.- Each cargo hook submitted for acceptance under a contract or purchase order shall have been successfully subjected to examination (4.6.1) and the operational load test (4.6.2.1) as described under 4.6.

4 5 Test Conditions Unless otherwise specified, tests shall be performed at prevailing ambient conditions with electric power specified in 3.9.7.1. The hook shall be rigidly, or semi-rigidly, mounted, as defined in paragraph 6.3, in a suitable test structure and the load applied using a nylon sling, similar to currently available items, with a minimum length of four feet. The test installation and methods shall simulate service conditions insofar as possible and shall be approved by the procuring agency prior to initiation of testing.

4.5.1 Order of Testing.- Unless otherwise specified, the order of testing shall be as specified herein. If the manufacturer desires a different sequence of testing or a combination of tests, he shall obtain prior approval from the procuring agency.

4.6 Tests.-

4.6.1 Examination of Product.- The cargo hook, its subassemblies and components shall be examined to determine conformance to this Specification and detail specifications with respect to materials, workmanship, and markings.

4.6.2 Test Procedures.-

4.6.2.1 Operational Load Test.- The operational load test shall consist of a specified number of cycles of operation. These cycles shall consist of:-

a. Engagement of load.
b. Hold load for two (2) minutes.
c. Release load with releases indicated below.
d. Relatch automatically.

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The following loads and release methods shall be used for the indicated number of cycles:-

No. of Cycles	Load	Release Method
10	6,000 lbs.	Normal
15	20,000 lbs.	Automatic Touchdown
5	7 lbs.	Normal
5	7 lbs.	Ground
2	20,000 lbs.	Manual
5	5,000 lbs.	Manual

The forces required to actuate the manual release shall be checked for no-load and full-load conditions after not less than 50 percent of each of the above tests have been completed. The release forces shall not exceed the values specified in paragraph 3.9.6.1.3.

4.6.2.2 Overload Test.- A static test load of 60,000 pounds shall be suspended from the hook for one (1) minute. The hook shall suffer no yield or damage, and shall operate satisfactorily through two (2) cycles of each of the tests listed in 4.6.2.1.

4.6.2.3 Overload Release Test.- The cargo hook manual release shall be actuated one (1) time with a 60,000 pound load suspended from the hook. The hook shall then be examined to see that there is no major damage and to ascertain that the hook can be returned to operating condition with only minor repair and replacement of minor parts. After examination and return of the hook to its normal operating condition, the hook shall operate satisfactorily.

4.6.2.4 Torque Test.- A load equal in magnitude to the rated operating capacity of the hook shall be suspended from it using a six (6) inch inside diameter steel ring, made from adequate diameter stock, to simulate a load pickup adapter ring. A torque of 10,000 inch-pounds shall then be applied to the ring for a minimum length of time of two (2) minutes. Upon completion of this requirement, the direction of the torque shall be reversed, and the above procedure followed again. The hook shall have suffered no permanent damage; and it shall operate satisfactorily through one (1) release of each of the tests listed in 4.6.2.1.

4.6.2.5 Angular Load Test.- The cargo hook shall be rigidly suspended from a rigid structure, and pitched up such that there is an angle of 30 degrees between the longitudinal axis of the hook and the horizontal. The hook shall then have vertical loads applied, gradually increasing from zero to rated capacity and returning to zero. The length of time for each loading cycle shall

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be a minimum of two minutes, and the number of cycles shall be thirty (30). Upon completion of the test, the hook shall be inspected for any permanent damage, and it shall be operated satisfactorily at this pitched attitude through one (1) cycle of each of the tests listed in paragraph 4.6.2.1.

4.6.2.6 Low Voltage Test.- A load of 6,000 pounds shall be suspended from the hook and the normal release circuit energized with an input voltage of 22 ± 1/2 VDC. The hook shall release the load and relatch automatically.

4.6.2.7 Maximum Current Drain.- Ammeters shall be installed in the input power supply and the hook shall be operated through three (3) normal releases while being subjected to a 6000-pound load with an input voltage of 28 ± 1/2 VDC. The maximum current drain shall not exceed 15 amperes.

4.6.2.8 Endurance Test.- The endurance test shall consist of the hook satisfactorily passing two complete series of cyclic tests performed in accordance with the chart below. Overhaul of the hook is permissible between the two series of tests. Under the overload (3 times rated load) condition, replacement of damaged minor parts is permissible in accordance with paragraph 4.6.2.3.

Cycles	Release Mode	Load
50	Normal	7 lbs.
1500	Automatic	10,000 lbs.
1	Manual	7 lbs.
500	Manual	1,000 lbs.
250	Manual	5,000 lbs.
1500	Normal	3,000 lbs.
750	Automatic	20,000 lbs.
750	Normal	6,000 lbs.
50	Manual	10,000 lbs.
25	Manual	20,000 lbs.
1	Manual	60,000 lbs.

Measurements shall be taken before and after the endurance test and a wear analysis shall be made showing the wear on each rubbing part. If during that portion of the test using the normal, or automatic touchdown mode, the hook fails to release, the manual release shall be actuated. If the hook then fails to release it shall be considered as an additional release failure. The number of release failures during the endurance tests on the hook shall not exceed two, not more than one of which shall be in the manual mode. During the endurance tests, the hook shall be allowed to relatch automatically. Not more than ten relatch failures shall be permitted during the complete endurance test. Failure

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to relatch at loads in excess of the rated capacity of the hook shall not constitute a failure. After any release failure in the manual mode, and subsequent to actuation of the manual release after a release failure in the normal mode, the hook may be examined to determine the cause of the failure. If the fault is determined for either a release or relatch failure, the deficiency shall be corrected. If the fault is not determined, the contractor may continue the tests. If the fault can be corrected by cleaning, adjustment, lubrication, or minor dressing of surfaces, the hook shall be reassembled and the testing continued. If corrective action requires general overhaul or replacement of parts, other than common fasteners, the testing shall be started from the beginning unless otherwise directed by the procuring agency. Should more than two release failures, or more than ten relatch failures occur during the endurance tests, the cause of failures shall be determined, adequate corrective action shall be taken and the testing started from the beginning. The forces required to actuate the manual release shall be checked for no-load, full-load and 60,000 pounds after each of the series of tests constituting the endurance tests. The forces shall not exceed the values indicated in para. 3.9.6.1.3.

4.6.2.9 Vibration Test.- The hook shall be subjected to a vibration test conforming to paragraph 4.7.12 (Procedure XII) of Specification MIL-E-5272. Upon completion of this test the hook shall be rigidly attached to a supporting fixture and the hook loaded. A vibrator, capable of producing resonant frequencies in the hook under all loads stated in the following chart, shall be attached to the fixture or the hook. All tests shall be conducted so as to produce a 2-1/2 ±1/2 g acceleration level at the hook. The hook shall be vibrated for the times, and under the loads, noted along each of its three mutually perpendicular axes. The noted times are totals and shall be equally divided among testing on each of the three axes. The tests shall consist of cycling vibration and resonant vibration with test time divided equally between them. The cycling vibration shall be conducted between 20 and 300 cycles per second and resonant frequencies in the hook shall be noted. Resonant vibration shall be performed at all of the major resonant points with equal time being used for each point when more than one resonant point is found. The hook release shall be actuated, as specified, while the hook is at a resonant point at the intervals, and under the loads, noted in the following chart:-

LOAD	TOTAL TIME	INTERVAL OF RELEASE	MODE OF RELEASE
7 lbs.	1/2 hour	Once on each axis of vibration	Normal
6,000 lbs.	6 hours	Three times on each axis of vibration *	Normal
20,000 lbs.	4 hours	One time on each axis of vibration **	Manual

* At least one release shall be made for each resonant point, up to three, if more than one resonant point is found.

** While at the major resonant point.

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No inadvertent releases shall be permitted. Additional tests shall be conducted at the discretion of the procuring agency if the results of the above tests indicate any questionable areas.

4.6.2.10 High Temperature Test.- The hook shall be tested as specified in paragraph 4.1.2 (Procedure II) of Specification MIL-E-5272 and the hook shall be tested through one (1) cycle of each of the tests listed in 4.6.2.1 herein.

4.6.2.11 Low Temperature Test.- The hook shall be tested as specified in paragraph 4.2.1 (Procedure I) of Specification MIL-E-5272 and the hook shall be tested through one (1) cycle of each of the tests listed in 4.6.2.1 herein.

4.6.2.12 Salt Spray Test.- The hook shall be tested as specified in paragraph 4.6.1 (Procedure I) of Specification MIL-E-5272 and, without any removal of any salt deposits, etc., resulting from the test, shall be operated satisfactorily at the conclusion of the test through one (1) cycle of each test condition listed in 4.6.2.1 herein.

4.6.2.13 Sand and Dust Test.- The hook shall be tested as specified in paragraph 4.11.1 (Procedure I) of Specification MIL-E-5272 except that the sand and dust velocity through the test chamber shall be maintained at 2500 ±500 feet per minute through the first half of the test and between 100 and 500 feet per minute through the second half of the test. Upon completion of the test, the hook shall be inspected for deterioration and shall be operated satisfactorily through one (1) cycle of each test condition in 4.6.2.1 herein.

4.6.2.14 Altitude Test.- The hook shall be tested as specified in paragraph 4.5.6 (Procedure VI) of Specification MIL-E-5272, except the maximum altitude shall be 20,000 feet. The hook shall be operated satisfactorily through one (1) cycle of each test condition listed in 4.6.2.1 herein, after subjection to altitude test.

4.6.2.15 Immersion Tests.- The hook shall be tested by immersion in sea water or water of equivalent salinity, to a depth of five (5) feet for fifteen (15) minutes and after being withdrawn, the hook shall be operated satisfactorily through one (1) cycle of each test condition listed in 4.6.2.1 herein.

4.6.2.16 Fungus Test.- The hook shall be tested as specified in paragraph 4.8.1 (Procedure I) of Specification MIL-E-5272, and, upon completion of this test, the hook shall be operated satisfactorily through one (1) cycle of each test condition listed in 4.6.2.1 herein.

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4.6.2.17 Shock Test.- The hook shall be tested in accordance with paragraph 4.15.5 (Procedure V) of Specification MIL-E-5272. The "equipment crash-safety" portion of this test shall be omitted. Upon completion of this test, a load shall be attached to the hook and dropped so as to induce a shock loading equivalent to 20,000 pounds at 3 g's. The hook shall successfully sustain ten (10) such shocks. Upon completion of this test the hook shall be operated satisfactorily through one (1) cycle of each test condition listed in 4.6.2.1 herein.

4.7 Re-examination and Retest.-

4.7.1 Retest.- The failure of the hook to meet the requirements of this specification shall cause tentative rejection subject to retest as provided herein. Any hook which fails the acceptance test may, upon the request of the manufacturer, be resubmitted for inspection and/or testing provided that a statement showing the corrective measures taken is furnished to the procuring agency at the time of such resubmission.

4.7.2 Second Retest.- If the failure of the retest is due to a remedial defect, the manufacturer may be offered the opportunity of again reworking the hook at the discretion of the procuring agency. Should the hook again fail, it shall be rejected.

4.7.3 Correction of Defects.- Any defects discovered during any inspection or testing shall be corrected by the manufacturer. Such corrective measures as may be found necessary shall likewise be incorporated on all hooks offered for acceptance under the contract or orders.

5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery.- Unless otherwise specified in the contract or purchase order, the cargo hook shall be prepared for delivery in accordance with Specification MIL-P-007936, unit packaging to be Level A and packing to be Level B and preservation shall be Method III of Specification MIL-P-116, utilizing a container conforming to Specification PPP-B-636.

6. NOTES

6.1 Intended Use.- The cargo hook is intended to provide a helicopter with an external means for cargo pickup, transport, and release when installed and connected to appropriate controls within the helicopter.

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6.2 Ordering Data.- Procurement documents should specify the following:-

(a) Title, number, and date of this Specification.
(b) Federal Stock Number or, if not available, the Manufacturer's Part Number.
(c) Handbook of Operating and Maintenance Instructions, or portions thereof, if required. (See 3.14).
(d) Disposition of qualification test sample if required. (See 4.3.1).
(e) Whether the contractor shall provide information for fulfillment of the requirements of Specification WR-30 (See 3.16). If required whether the information shall be supplied directly to the helicopter prime contractor or to the procuring activity.
(f) Type of preservation, packaging, and packing required if applicable. (See 5.1).

6.3 Definitions.-

6.3.1 Rigid Mount.- When the cargo hook is attached to the supporting structure so that there is no relative motion between them, it is a rigid mount.

6.3.2 Semi-Rigid Mount.- When the cargo hook is attached to a member of the supporting structure such that the hook and member can pivot, or swing, about a longitudinal axis located just above the hook, it is a semi-rigid mount.

6.4 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this Specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington, D.C. and information pertaining to qualification of products may be obtained from the Naval Air Development Center (Code EDE), Johnsville, Pennsylvania.

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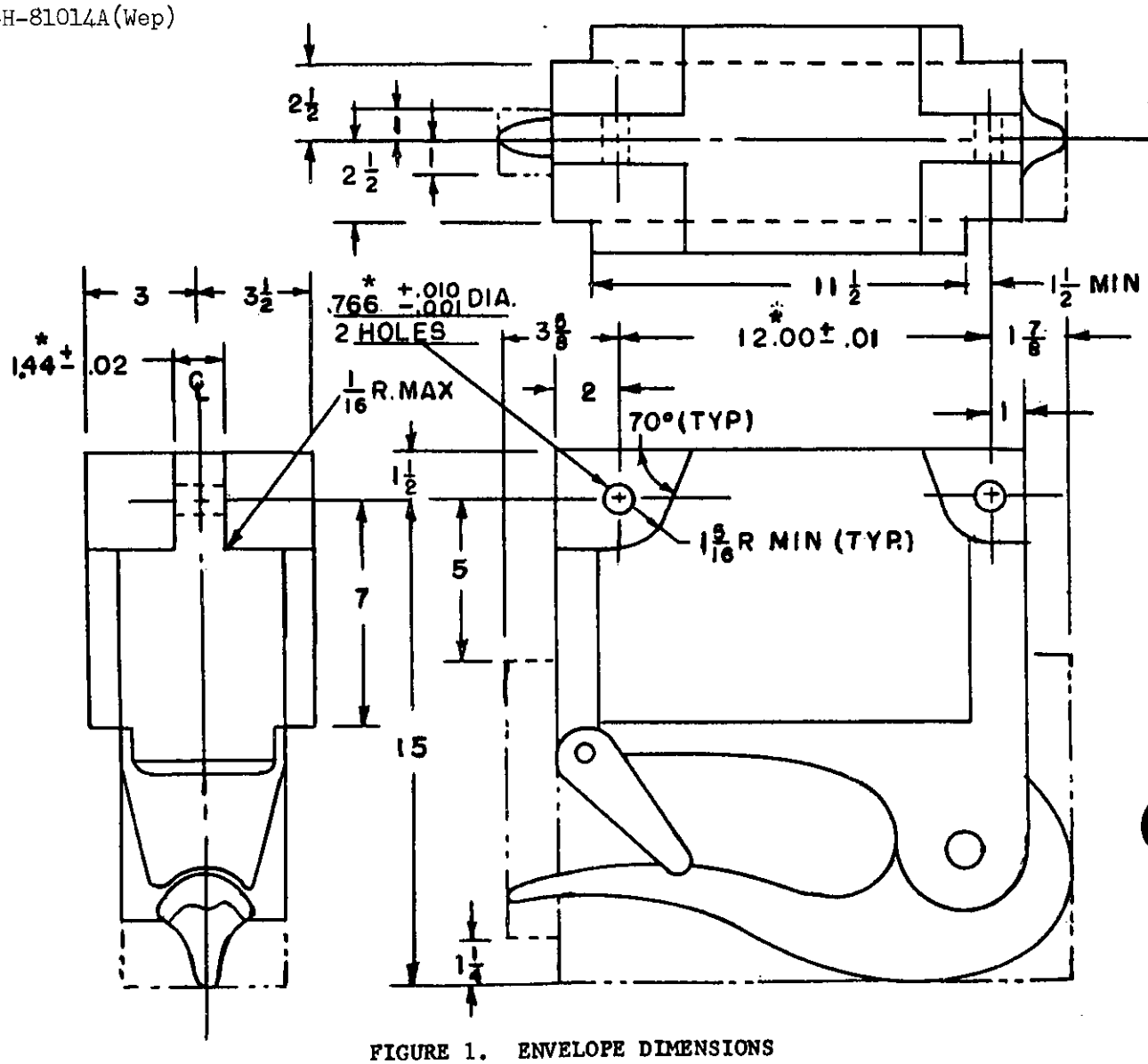


FIGURE 1. ENVELOPE DIMENSIONS

NOTES: 1. All dimensions marked with asterisk (*) must be held.

2. All other dimensions define maximum envelope size except as noted.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-H-81014A(WP) HOOK, HELICOPTER EXTERNAL CARGO, 20,000 POUND CAPACITY TYPE C

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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